

Computer Architectures

Exam of 18.2.2026 - part I

Question #1A

Let consider the mechanism of exceptions. You are requested to

1. List the events that can trigger an exception in a processor, grouping them in categories
2. Define what we mean by *precise exception* and why precision exception management is important
3. Describe how we can guarantee precise exception management in pipelined processors
4. Describe how we can guarantee precise exception management in superscalar processors with dynamic scheduling and speculation.

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Question #2A

Let consider the Branch Prediction Mechanism based on the Branch History Table (BHT).

You are requested to

1. Describe the architecture of a BHT
2. Describe the behavior of a BHT, detailing when the processor accesses it, how, and which information it gets from it

Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, the BHT is composed of 16 1-bit entries, and is initially empty (i.e., full of 0s, meaning NotTaken), consider the following instructions executed in sequence (for each instruction the corresponding address is reported)

- o 0x00B80250 lw x1, 0(x2)
- o 0x00B80254 bne x1, x10, lab1 (branch taken, jumping to 0x00008000)
- o 0x00008000 addi x2, x2, #1
- o 0x00008004 beq x2, x11, lab2 (branch not taken)
- o 0x00008008 andi x5, x5, #1
- o 0x0000800C beq x5, x12, lab3 (branch taken, jumping to 0x0000AA10)
- o 0x0000AA10 add x1, x2, x3
- o 0x0000AA14 beq x1, x13, lab4 (branch taken, jumping to 0x00004008)
- o 0x00004008 lw x6, 4(x2)
- o 0x0000400C addi x2, x2, #1
- o 0x00004010 bne x2, x10, lab5 (branch taken, jumping to 0x00009200)
- o 0x00009200 add x4, x5, x0
- o 0x00009204 beq x4, x11, lab6 (branch taken, jumping to 0x0000AAF0)
- o 0x0000AAF0 sw x4, 8(x2)
- o 0x0000AAF4 beq x4, x12, lab1 (branch taken, jumping to 0x00008000)

Using the following tables, you are asked to

3. Determine which of the branch instructions were correctly predicted, and which were mispredicted
4. Identify the final content of the BHT.

Address	Instruction	Addressed BHT line	Prediction (T/NT)	Result (T/NT)	Prediction (correct/wrong)
0x00B80250	lw x1, 0(x2)				
0x00B80254	bne x1, x10, lab1				
0x00008000	addi x2, x2, #1				
0x00008004	beq x2, x11, lab2				
0x00008008	andi x5, x5, #1				
0x0000800C	beq x5, x12, lab3				
0x0000AA10	add x1, x2, x3				
0x0000AA14	beq x1, x13, lab4				
0x00004008	lw x6, 4(x2)				
0x0000400C	addi x2, x2, #1				
0x00004010	bne x2, x10, lab5				
0x00009200	add x4, x5, x0				
0x00009204	beq x4, x11, lab6				
0x0000AAF0	sw x4, 8(x2)				
0x0000AAF4	beq x4, x12, lab1				

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Initial BHT content	
Index	Content
0	0
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0

Final BHT content	
Index	Content
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	